Air pollution, Solid & Water pollution, CO_2 emissions, AQI and Government regulations, Emission Factor

AIR POLLUTION

Air pollution can be defined as "an alteration of air quality that may be characterised by measurement of chemical, biological and/or physical pollutants in the air". Air pollution occurs when harmful or excessive quantities of substances are introduced into Earth's atmosphere. Sources of air pollution include gases (such as ammonia, carbon monoxide, sulphur dioxide, nitrous oxides, methane and chlorofluorocarbons), particulates (both organic and inorganic), and biological molecules. It may cause diseases, allergies and even death to humans; it may also cause harm to other living organisms such as animals and food crops, and may damage the natural or built environment. Both human activity and natural processes can generate air pollution.

Air pollution can be divided area wise as follows:

- Local: concerns the quality of ambient air within a radius of like, a few kilometres
- Regional: includes pollution such as acid rain, photochemical reactions and degradation of water quality at distances of a few kilometres to a thousand kilometres
- Global: depletion of the ozone layer and global warming caused by emission of greenhouse gases (mainly carbon dioxide [CO2])

Types of pollutants

Pollutants that directly form from human or natural activities are called primary pollutants. Main primary pollutants are

- **Carbon dioxide (CO2)** – Because of its role as a greenhouse gas it has been described as "the leading pollutant" and "the worst climate pollutant". Carbon dioxide is a natural component of the atmosphere, essential for plant life and given off by the human respiratory system. CO2 increase in earth's atmosphere has been accelerating.
- **Sulphur oxides (SOx)** – particularly sulphur dioxide, a chemical compound with the formula SO2. SO2 is produced by volcanoes and in various industrial processes. Coal and petroleum often contain sulphur compounds, and their combustion generates sulphur dioxide.
- **Nitrogen oxides (NOx)** – Nitrogen oxides, particularly nitrogen dioxide, are expelled from high temperature combustion, and are also produced during thunderstorms by electric discharge. They can be seen as a brown haze dome above or a plume downwind of cities.
- **Carbon monoxide (CO)** – CO is a colourless, odourless, toxic gas. It is a product of combustion of fuel such as natural gas, coal or wood. Vehicular exhaust contributes to the majority of carbon monoxide let into our atmosphere.
- **Volatile organic compounds (VOC)** – VOCs are a well-known outdoor air pollutant. They are categorized as either methane (CH4) or non-methane (NMVOCs). Methane is an extremely efficient greenhouse gas which contributes to enhanced global warming.
- **Particles in air** are either: directly emitted, for instance when fuel is burnt and when dust is carried by wind, or indirectly formed, when gaseous pollutants previously emitted to air turn into particulate matter

  Based on size, Particulate matter is often divided into two main groups:
  1. The coarse fraction contains the larger particles with a size ranging from 2.5 to 10 µm (PM10 – PM2.5).

2. The fine fraction contains the smaller ones with a size up to (PM2.5 μm). The particles in the fine fraction which are smaller than 0.1 μm are called ultra-fine particles.

Most of the total mass of airborne Particulate matter is usually made up of fine particles ranging from 0.1 to 2.5 μm. Ultra-fine particles often contribute only a few percent to the total mass, though they are the most numerous, representing over 90% of the number of particles

Secondary pollutants are formed when primary pollutants interact with each other. Examples are

- Particulates created from gaseous primary pollutants and compounds in photochemical smog. Smog is a kind of air pollution. Classic smog results from large amounts of coal burning in an area caused by a mixture of smoke and sulphur dioxide. Modern smog does not usually come from coal but from vehicular and industrial emissions that are acted on in the atmosphere by ultraviolet light from the sun to form secondary pollutants that also combine with the primary emissions to form photochemical smog.
- Ground level ozone (O3) formed from NOx and VOCs. Ozone (O3) is a key constituent of the troposphere. It is also an important constituent of certain regions of the stratosphere commonly known as the Ozone layer.
- Peroxyacetyl nitrate (C2H3NO5) – similarly formed from NOx and VOCs.

Causes of Air Pollution

- **The burning of fossil fuels:** Sulphur dioxide emitted from the combustion of fossil fuels such as coal, petroleum and in factories is one the major causes of air pollution. Emissions from vehicles including trucks, jeeps, cars, trains and airplanes cause an immense amount of pollution. The overuse of these vehicles kills our environment as dangerous gases accumulated. Carbon Monoxide released by improper (or incomplete) combustion in vehicles is another major pollutant along with Nitrogen Oxides, which are produced from both natural and man-made processes.
- **Agricultural activities:** Ammonia is a very common by-product of agriculture-related activities and is one of the most hazardous gases in the atmosphere. Use of insecticides, pesticides, and fertilizers in agricultural activities emit harmful chemicals into the air and can also degrade water. Methane and nitrous oxide were the two main gases emitted by agricultural activity.
- **Exhaust from factories and industries:** Manufacturing industries release a large amount of carbon monoxide, hydrocarbons, organic compounds and chemicals into the air thereby depleting its quality. Manufacturing industries can be found at every corner of the earth and there is no area that has not been affected by it. Petroleum refineries also release hydrocarbons and various other chemicals that pollute the air and also cause land pollution.
- **Mining operations**: Mining is a process wherein minerals below the earth are extracted using heavy equipment. In the mining process a lot of activities are involved: drilling, blasting, hauling, collection, and transportation, releasing a lot of dust into the air. Air pollution from coal mines is particularly harmful due to emissions of particulate matter and gases, including methane, sulphur dioxide, and oxides of nitrogen, and carbon monoxide.
- **Indoor air pollution:** Household cleaning products, painting equipment etc. emit toxic chemicals into the air. Cooking and heating with polluting fuels and technologies produces high levels of household air pollution which includes a range of health damaging pollutants such as fine particles and carbon monoxide.

Disastrous Effects of Air pollution

- **Respiratory and heart problems**: increased air pollution creates several respiratory and other health threats to the body. Particularly small children in areas exposed to air pollutants are said to commonly suffer from pneumonia and asthma. Air pollution is known to cause irritation in the eyes, lungs, nose, and throat. When continually exposed to air pollution, humans become at higher risk for cardiovascular disease. Air filled with toxins can have a number of adverse effects on the arteries, and have even been a contributor to heart attacks.
- **Global Warming**: another direct effect of air pollution. With increased temperatures worldwide, increase in sea levels, melting of ice from colder regions and icebergs, and displacement and loss of habitat have already signalled an impending disaster if actions for preservation and normalisation aren’t undertaken soon. Air pollution directly accelerates the rate at which global warming happens by depleting the Ozone layer.
- **Acid Rain**: harmful gases like Nitrogen oxides and Sulphur oxides are released into the atmosphere during the burning of fossil fuels. When it rains, the water droplets combine with these air pollutants, become acidic and then falls on the ground in the form of acid rain. Acid rain can cause great damage to human, animal and plant lives.
- **Eutrophication:** eutrophication is a condition where a high amount of nitrogen present in some pollutants gets developed on water surfaces and turns itself into algae which adversely affects fish, plant and animal species. The most conspicuous effect of cultural eutrophication is the creation of dense blooms of noxious, foul-smelling phytoplankton that reduce water clarity and harm water quality.
- **Effect on Wildlife:** just like humans, animals also face some devastating effects of air pollution. Toxic chemicals present in the air can force wildlife species to migrate and change their habitat. The toxic pollutant-deposits over the water surfaces can affect aquatic life. Heavily polluted areas force inhabitants to seek new homes, which can negatively impact the ecosystem. Toxic chemicals also deposit over surfaces of water that can lead to the endangerment of marine life animals.
- **Depletion of the ozone layer:** ozone exists in the Earth’s stratosphere and is responsible for protecting humans from harmful ultraviolet (UV) rays. Earth’s ozone layer is depleting due to the presence of chlorofluorocarbons and hydro chlorofluorocarbons in the atmosphere. As the ozone layer goes thin, it will emit harmful rays on to the earth and can cause skin and eye related problems. UV rays also affect crops.

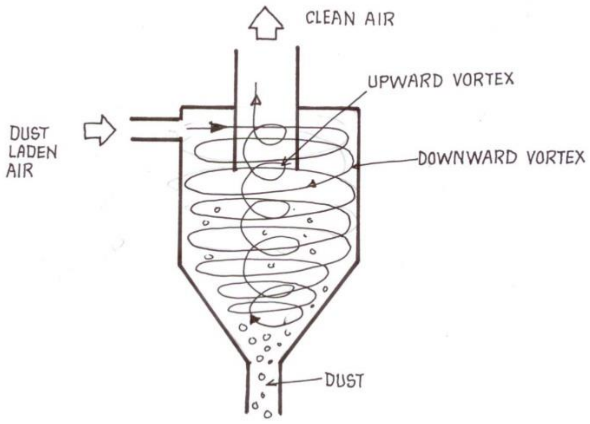
Control of Air pollution

- The main source of air pollution is automobiles. So, the engines of automobiles should be redesigned in such a way that their emissions cause minimal pollution. The vehicle should be serviced regularly. The traditional use of wood and coal as fuel should be checked and newly devised smoke free furnaces should be used.
- Forest fires need immediate measures as and when it happens. Forest covers must be protected. 33% area of land requires to be covered by the forests. The cutting of trees should be controlled. The reforestation i.e. planting more trees shall be promoted. The forests are considered as the lungs of atmosphere.
- The Green Belts around the roads, residential areas as well as industries are necessary.
- Burning of Crackers on various occasions should be banned and people should be sensitized about the ill effects produced by these.
- The stubble burning in agricultural farms must be controlled.
- Plant more trees and take proper care of plants. EACH ONE PLANT ONE should be promoted.

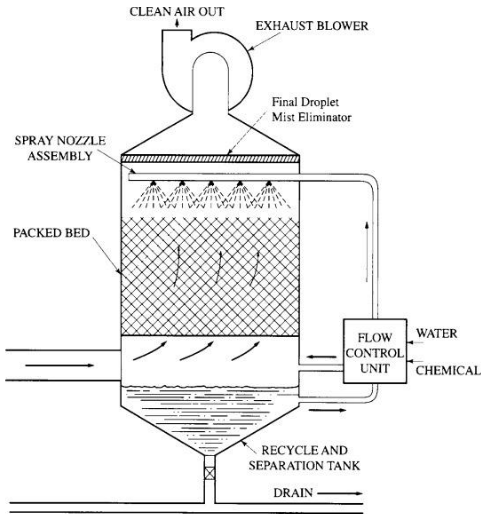
Control of Air pollution in Industries

Control of Particulates

- Cyclonic separation: Cyclonic separation removes particulates by causing the dirty airstream to flow in a spiral path inside a cylindrical chamber. Dirty air enters the chamber from a tangential direction at the outer wall of the device, forming a vortex as it swirls within the chamber. The larger particulates, because of their greater inertia, move outward and are forced against the chamber wall. Slowed by friction with the wall surface, they then slide down the wall into a conical dust hopper at the bottom of the cyclone.



- Scrubber: Devices called wet scrubbers trap suspended particles by direct contact with a spray of water or other liquid. In effect, a scrubber washes the particulates out of the dirty airstream as they collide with and are entrained by the countless tiny droplets in the spray.



- Electrostatic precipitators: Electrostatic precipitation is a commonly used method for removing fine particulates from airstreams. In an electrostatic precipitator, particles suspended in the airstream are given an electric charge as they enter the unit and are then removed by the influence of an electric field. The precipitation unit comprises baffles for distributing airflow, discharge and collection electrodes, a dust clean-out system, and collection hoppers. A high voltage of direct current (DC), as much as 100,000 volts, is applied to the discharge electrodes to charge the particles, which then are attracted to oppositely charged collection electrodes, on which they become trapped.

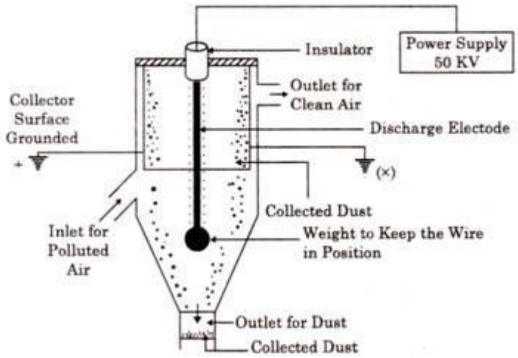
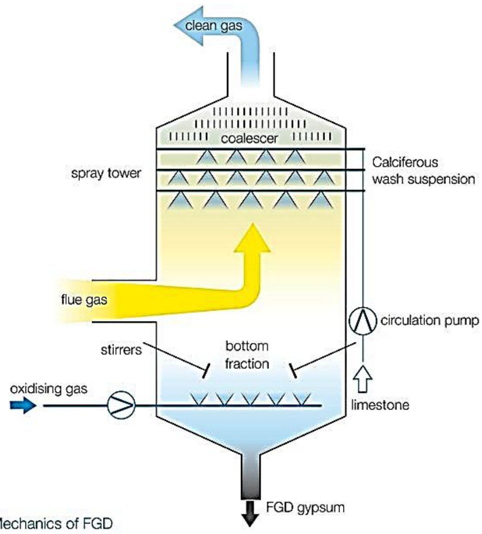


Fig. 5.4 Electrostatic Precipitator

Control of Gases

- Flue gas desulphurisation (FGD): FGD systems may involve wet scrubbing or dry scrubbing. In wet FGD systems, flue gases are brought in contact with an absorbent, which can be either a liquid or a slurry of solid material. The sulphur dioxide dissolves in or reacts with the absorbent and becomes trapped in it. In dry FGD systems, the absorbent is dry pulverized lime or limestone; once absorption occurs, the solid particles are removed by means of baghouse filters (described above). Dry FGD systems, compared with wet systems, offer cost and energy savings and easier operation, but they require higher chemical consumption and are limited to flue gases derived from the combustion of low-sulphur coal.



- Incineration: The process called incineration or combustion chemically, rapid oxidation can be used to convert VOCs and other gaseous hydrocarbon pollutants to carbon dioxide and water. Incineration of VOCs and hydrocarbon fumes usually is accomplished in a special incinerator called an afterburner. To achieve complete combustion, the afterburner must provide the proper amount of turbulence and burning time, and it must maintain a sufficiently high temperature. Sufficient turbulence, or mixing, is a key factor in combustion because it reduces the required burning time and temperature. A process called direct flame incineration can be used when the waste gas is itself a combustible mixture and does not need the addition of air or fuel.

Major Air Pollution Disasters in the world

- The very first case of air pollution disaster was in the Meuse Valley of Belgium in 1930. The **1930 Meuse Valley fog** killed 60 people in Belgium due to a combination of industrial air pollution and climatic conditions in December that year. The Meuse river flows from France through Belgium and the Netherlands before entering the North Sea. The area in the Meuse Valley where the incident occurred is densely populated as well as having many factories. There were several thousand cases of illness over the period of two or three days with the sixty deaths occurring at the same time. Fifty-six of the deaths were to the east of Engis. The main symptom was dyspnoea (shortness of breath) and the average age of those who died was 62, over a range of ages of 20 to 89 years. Cattle in the area were also affected.

- Another case was a Killing Smog in Donora, Pennsylvania in 1948 in which hundreds of people died. The 1948 Donora smog was a historic smog event that killed 20 people and caused respiratory problems for 6,000 people of the 14,000 population of Donora, Pennsylvania, a mill town on the Monongahela River 24 miles (39 km) southeast of Pittsburgh.
- A Deadlier **Smog in London** in 1952, in which about 4000-5000 people died from respiratory failure is another air pollution episode. The mixture of smoke, fog and oxides of sulphur affected London so badly for several centuries after the introduction of coal as a fuel.
- The **Bhopal disaster**, also referred to as the Bhopal gas tragedy, was a gas leak incident on the night of 2–3 December 1984 at the Union Carbide India Limited (UCIL) pesticide plant in Bhopal, Madhya Pradesh, India. It is considered to be the world's worst industrial disaster. Over 500,000 people were exposed to methyl isocyanate (MIC) gas. The highly toxic substance made its way into and around the small towns located near the plant. A government affidavit in 2006 stated that the leak caused 558,125 injuries, including 38,478 temporary partial injuries and approximately 3,900 severely and permanently disabling injuries. Others estimate that 8,000 died within two weeks, and another 8,000 or more have since died from gas-related diseases.

WATER POLLUTION

Water pollution is the contamination of water bodies, usually as a result of human activities. Water bodies include lakes, rivers, oceans, aquifers and groundwater. Water pollution results when contaminants are introduced into the natural environment. For example, releasing inadequately treated wastewater into natural water bodies. Sources of water pollution are either point sources and non-point sources. Point sources have one identifiable cause of the pollution, such as a storm drain, wastewater treatment plant or stream. Water is an easy solvent, enabling most pollutants to dissolve in it easily and contaminate it. The most basic effect of water pollution is directly suffered by the organisms and vegetation that survive in water.

Water is typically referred to as polluted when it is impaired by anthropogenic contaminants. Due to these contaminants it either does not support a human use, such as drinking water, or undergoes a marked shift in its ability to support its biotic communities, such as fish. Natural phenomena such as volcanoes, algae blooms, storms, and earthquakes also cause major changes in water quality and the ecological status of water.

TYPES OF WATER POLLUTION

1. CHEMICAL WATER POLLUTION.

Chemical pollution is the presence or increase in our environment of chemical pollutants that are not naturally present there or are found in amounts higher than their natural background values. Chemical compounds are of 2 types organic or inorganic chemicals which are the main causes of chemical pollution. Examples: pesticides, herbicides, insecticides used in agriculture and gardening.

2. OIL SPILLAGE.

An oil spill is the release of a liquid petroleum hydrocarbon into the environment. Oil spills penetrate into the structure of the plumage of birds and the fur of mammals, reducing its insulating ability, and making them more vulnerable to temperature fluctuations and much less buoyant in the water. It affects the ecosystem of the ocean.

3. SUSPENDED MATTER.

Suspended solids refer to small solid particles which remain in suspension in water as a colloid of suspended solids. The suspended particles eventually settle and cause a thick silt at the bottom. This is harmful to marine life that lives. Toxic chemicals suspended in water can also be harmful to the development and survival of aquatic life.

4. GROUNDWATER POLLUTION.

Groundwater contamination occurs when man-made products such as gasoline, oil, road salts and chemicals get into the groundwater and cause it to become unsafe for human use. Materials from the land's surface can move through the soil to the groundwater. Example, pesticides and fertilizers.

5. NUTRIENTS POLLUTION.

Nutrient pollution is the process where many nutrients, mainly nitrogen and phosphorus, causing excessive growth of algae. This process is also known as eutrophication. Excessive amounts of nutrients can lead to low levels of oxygen dissolved in the water. The oxygen in the water is used up and this leads to low levels of dissolved oxygen in the water which affects the aquatic life.

SOURCES OF WATER POLLUTION.

Water is one of the renewable resources essential for sustaining all forms of life, food production, economic development, and for general well-being. Water pollution results when contaminants are introduced into the natural environment. Water pollution is a major environmental issue in India. The largest source of water pollution in India is untreated sewage. Other sources of pollution include agricultural runoff and unregulated small-scale industry. Most rivers, lakes and surface water in India are polluted due to industries, untreated sewage and solid wastes.

CAUSES OF WATER POLLUTION.

1. INDUSTRIAL MINING

Industries produce a huge amount of waste which contains toxic chemicals and pollutants which can cause water pollution. They contain pollutants such as lead, mercury, sulfur, asbestos, nitrates, and many other harmful chemicals, such toxic chemicals have the capability to change the color of water, eutrophication, change in temperature of water.

2. SEWAGE AND WASTEWATER

The sewage water carries harmful bacteria and chemicals that can cause serious health problems. Pathogens are a common water pollutant. Microorganisms in water are known to be causes of some very deadly diseases and become the breeding grounds for other creatures that act as carriers.

3. MINING ACTIVITIES

The elements when extracted in the raw form contains harmful chemicals and can increase the number of toxic elements when mixed up with water which may result in health problems. Mining activities emit a large amount of metal waste and sulphides from the rocks which is harmful to the water

4. RADIOACTIVE WASTE

The element is used in the production of nuclear energy is Uranium which is a highly toxic chemical. The nuclear waste needs to be disposed of to prevent any nuclear accident. Nuclear waste can have serious environmental hazards if not disposed of properly. The most currently-used method for nuclear waste disposal is storage, using steel cylinders as radioactive shield or using deep and stable geologic formations.

MEASUREMENT OF WATER QUALITY

Water quality testing is important part of environmental monitoring. When water quality is poor, it affects not only aquatic life but the surrounding ecosystem as well. Physical properties of water quality include temperature and turbidity. Chemical characteristics involve parameters such as pH and dissolved oxygen. Some tests require the measurement to be conducted in the stream since the process of obtaining a sample may change the measurement. The measurement should be conducted right in the stream, or the sample must be extracted with great care to ensure that no transfer of oxygen from the air and water (in or out) occurs.

1. DISSOLVED OXYGEN

The most important measure of water quality is the dissolved oxygen (DO). Oxygen, although poorly soluble in water, is fundamental to aquatic life. Without free DO, streams and lakes become uninhabitable to gill-breathing aquatic organisms. Dissolved oxygen is inversely proportional to temperature, and the maximum oxygen that can be dissolved in water at most ambient temperature is about 10 mg/L. The amount of oxygen dissolved in water is usually measured either with an oxygen probe or by iodometric titration. The principle of operation of an oxygen probe is that of a galvanic cell. If lead and silver electrodes are put into an electrolyte solution with a micro ammeter between, the reaction at the lead electrode will be

$Pb + 2OH^- \rightarrow PbO + H_2O + 2e$,At the lead electrode (the cathode), electrons are liberated that travel through the micro ammeter to the silver electrode (the anode), where the following re- action takes place: $2e^- + \frac{1}{2} O_2 + H_2O \rightarrow 2OH^-$ The overall reaction, balancing the electrons, is then

$Pb + \frac{1}{2} O_2 \rightarrow PbO$. The electricity recorded is proportional to the concentration of oxygen in the electrolyte solution.

2. PH

The pH of a solution is a measure of hydrogen ion concentration, which in turn is a measure of its acidity. Pure water dissociates slightly into equal concentrations of hydrogen and hydroxyl (OH^-) ions.

An excess of hydrogen ions makes a solution acidic, whereas a dearth of H^+ ions, or an excess of hydroxyl ions, makes it basic. The equilibrium constant for this reaction, Kw, is the product of H^+ and OH^- concentrations and is equal to 10^{-14}. This relationship may be expressed as

$[H^+][OH^-] = K_w = 10^{-14}$ In pure water,

$[H^+] = [OH^-] = 10^{-7}$ moles/L

The measurement of pH is now almost universally by electronic means. Electrodes that are sensitive to hydrogen ion concentration (strictly speaking, the hydrogen ion activity) convert the signal to electric current, pH is important in almost all phases of water and wastewater treatment. Aquatic organisms are sensitive to pH changes, and biological treatment requires either pH control or monitoring. In water treatment as well as in disinfection and corrosion control, pH is important in ensuring proper chemical treatment.

2. TURBIDITY, TSS AND CLARITY

The solar radiation provides light, heat, and energy to all living being on earth. Suspended solids, decaying vegetation, and other dissolved material cause the water to appear cloudy and murky, impacting the penetration of sunlight on water and the aquatic life. A sudden increase in the turbidity and the total suspended solids (TSS) is an indicator of soil erosion and point-source pollution adding heavy metals and effluents into the water. A nephelometer is used to measure the scattered light at an angle of 90◦ and the results are reported in Nephelometric Turbidity Unit (NTU). The total suspended solids are measured by filtering and weighing the sample and are measured in milligrams of solids per litre of water.

3. CDOM/FDOM MONITORING

Coloured or chromophoric dissolved organic matter (CDOM) occurs naturally in water bodies. This organic matter absorbs the ultraviolet light and decomposes to release tannin, an organic pollutant that causes the water to turn murky. Moreover, tannin contributes to reducing the pH (acidic) of the water and depleting the oxygen levels. CDOM/FDOM levels can be measured using electrical optical sensors that use fluorometers and sapphire lens.

4. NITRATES

Nitrates are a form of nitrogen, which is found in several different forms in terrestrial and aquatic ecosystems. These forms of nitrogen include ammonia (NH_3), nitrates (NO_3), and nitrites (NO_2). Nitrates are essential plant nutrients, but in excess amounts they can cause significant water quality problems. Together with phosphorus, accelerate eutrophication, causing dramatic increases in aquatic plant growth and changes in the types of plants and animals that live in the stream. This, affects dissolved oxygen, temperature, and other indicators. The natural level of ammonia or nitrate in surface water is typically low (less than 1 mg/L); in the effluent of wastewater treatment plants, it can range up to 30 mg/L.

5. COD & BOD OF WATER

A **chemical oxygen demand (COD)** test is used to measure the amount of organic compounds in a water sample. It measures the capacity of water to consume oxygen during the decomposition of organic matter and the oxidation of inorganic chemicals such as Ammonia and nitrate. It is expressed in mass of oxygen consumed over volume of solution which in SI units is milligrams per litre (mg/L).

MEASUREMENT: For all organic matter to be completely oxidized, an excess amount of potassium dichromate (or any oxidizing agent) must be present. Once oxidation is complete, the amount of excess potassium dichromate must be measured to ensure that the amount of $Cr3+$ can be determined with accuracy. To do so, the excess potassium dichromate is titrated with ferrous ammonium sulphateS (FAS) until all of the excess oxidizing agent has been reduced to $Cr3+$. Once all the excess dichromate has been reduced, the ferroin indicator changes from blue-green to a reddish brown. During the titration the colour of the indicator changes from a green hue to a bright blue hue to a reddish brown upon reaching the endpoint.

Biochemical oxygen demand (BOD), is a chemical procedure for determining the amount of dissolved oxygen needed by aerobic biological organisms in a body of water to break down organic material present in a given water sample at certain temperature. It is most commonly expressed in milligrams of oxygen consumed per litre of sample during 5 days (BOD_5) of incubation at 20°C. The greater the value, the more rapidly oxygen is depleted in the stream. This means less oxygen is available to higher forms of aquatic life.

Manometric method (BOD): This method is limited to the measurement of the oxygen consumption due only to carbonaceous oxidation. Ammonia oxidation is inhibited. The sample is kept in a sealed container fitted with a pressure sensor. A substance that absorbs carbon dioxide (typically lithium hydroxide) is added in the container above the sample level. Oxygen is consumed and, as ammonia oxidation is inhibited, carbon dioxide is released. The total amount of gas, and thus the pressure, decreases because carbon dioxide is absorbed. From the drop of pressure, the sensor electronics computes and displays the consumed quantity of oxygen.

WATER POLLUTION CONTROL

Control of water pollution requires appropriate infrastructure and management plans. The infrastructure may include wastewater treatment plants. Sewage treatment plants and industrial wastewater treatment plants are usually required to protect water bodies from untreated wastewater. Agricultural wastewater treatment for farms, and erosion control from construction sites can also help prevent water pollution. Effective control of urban runoff includes reducing speed and quantity of flow.

1. MUNICIPAL WASTEWATER TREATMENT

Well-designed and operated systems (i.e., with secondary treatment steps or more advanced treatment) can remove 90 percent or more of the pollutant load in sewage. It mainly includes Phase separation (remove solids generated during oxidation or polishing), Biochemical oxidation, polishing (chemical reduction and PH adjustments).

2. ON-SITE SANITATION AND SAFELY MANAGED SANITATION

Households or businesses that are not served by a municipal treatment plant may have an individual septic tank, which pre-treats the wastewater on site and infiltrates it into the soil. The purpose of on-site wastewater treatment systems is to reduce the concentrations of contaminants. This can lead to groundwater pollution if not properly done.

3. INDUSTRIAL WASTEWATER TREATMENT

Some industries install a pre-treatment system to remove some pollutants (e.g., toxic compounds), and then discharge the partially treated wastewater to the municipal sewer system. reduce or eliminate pollutants, through a process called pollution prevention.

4. AGRICULTURAL WASTEWATER TREATMENT

A farm management agenda for controlling pollution from surface runoff that may be contaminated by chemicals in fertilizer, pesticides, animal slurry, crop residues or irrigation water. It has 2 types the point source pollution and the non-point pollution. Farmers may utilize erosion controls to reduce runoff flows and retain soil on their fields. Common techniques include contour ploughing, crop mulching, crop rotation, planting perennial crops and installing riparian buffers.

5. EROSION AND SEDIMENT CONTROL FROM CONSTRUCTION SITES

Sediment control limits the amount of sediment that is carried into lakes, streams and rivers by storm water runoff. A sediment control is a practice or device designed to keep eroded soil on a construction site, so that it does not wash off and cause water pollution to a nearby stream, river, lake, or sea. They are designed to prevent or minimize erosion and thus reduce the need for sediment controls. Such as check dam, diversion dike, Fibre roll.

GLOBAL AND INDIAN SCENARIO OF WATER POLLUTON

Water pollution is on the rise globally. Virtually all goods-producing activities generate pollutants as unwanted by-products. Water pollution can be grouped into surface water pollution. Marine pollution and nutrient pollution are subsets of water pollution. Sources of water pollution are either point sources and non-point sources. Point sources have one identifiable cause of the pollution, such as a storm drain, wastewater treatment plant or stream. Projected food production needs and increasing wastewater effluents associated with an increasing population over the next three decades suggest a 10%-15% increase in the river input of nitrogen loads into coastal ecosystems, continuing the trend observed during 1970-95. More than 80% of sewage in developing countries is discharged untreated, polluting rivers, lakes and coastal areas. Despite improvements in some regions, water pollution is on the rise globally. Water pollution is a major environmental issue in India. The largest source of water pollution in India is untreated sewage. Other sources of pollution include agricultural runoff and unregulated small-scale industry. Most rivers, lakes and surface water in India are polluted due to industries, untreated sewage and solid wastes. The majority of the government-owned sewage treatment plants remain closed most of the time due to improper design or poor maintenance or lack of reliable electricity supply to operate the plants, together with absentee employees and poor management. The waste water generated in these areas normally percolates into the soil or evaporates. The uncollected waste accumulates in the urban areas causing unhygienic conditions and releasing pollutants that leach into surface and groundwater.

CASE STUDY

Pollution in Ganges

More than 500 million people live along the Ganges river. An estimated 2,000,000 persons ritually bath daily in the river, which is considered holy by Hindus. Ganges river pollution is a major health risk. NRGBA was established by the central government of India, on 20 February 2009 under section 3(3) of the environment protection act, 1986. It also declared Ganges as the "national river" of India. The chair includes the prime minister of India and chief ministers of states through which the Ganges flows. Today, the ganga is considered to be the sixth-most polluted river in the world. The Namami Gange project was announced by the government in the June 2014 budget. An estimated Rs. 2,958 crores (us$460 million) have been spent until July 2016 in various efforts in cleaning up of the river.

Pollution in Yamuna

In 1909, the waters of the Yamuna were distinguishable as clear blue, when compared to the silt-laden yellow of the Ganges. However, due to high-density population growth and rapid industrialization, Yamuna has become one of the most polluted rivers in India. The Yamuna is particularly polluted downstream of New Delhi, the capital of India, which dumps about 58% of its waste into the river. Lack of sufficient number of sewage treatment plants has led to the increase in the polluted stretch of the Yamuna. Yamuna Action Plan (YAP) – Yamuna Action Plan is for cleaning the Yamuna. Since 1993 Japan International Cooperation Agency, Government of Japan is assisting the Government of India to clean the Yamuna in phases. 39 sewage treatment plants in 29 towns of Uttar Pradesh, Haryana and Delhi were built in phase I of the plan. Near about Rs1,500 crores have been spent under Yamuna Action Plan I and II.

SOLID POLLUTION

Solid wastes are all the discarded solid materials from municipal, industrial, and agricultural activities. Solid waste is the 3rd largest polluter after air and water. It includes bottles, crockery, plastics, polythene and other packing materials etc. The objective of solid wastes management (SWM) are to control, collect, process, utilize and dispose of solid wastes in an economical way consistent with the public health protection.

WHAT IS SOLID WASTE?

- Solid waste refers to any garbage, refuge, sludge or other discarded materials including solid, liquid, semi solid or contained gaseous material, resulting from industrial and commercial activities mainly
- Does not include solid or dissolved materials or discharges.

SOURCES

- Wastes are generated from agriculture, mining and oil and gas production (over 85%)
- Less than 15% of the generated waste comes from industrial and municipal sources
- In Canada and the USA, the average citizen produces about 700 to 850 kgs of waste per year which is more than twice the quantity generated by individuals living in Japan
- In developing countries, the average quantity of waste per person is between 100 and 200 kgs

Characterization of Solid Waste

- Nature – Organic, Inorganic, Putrescible
- Combustible, Recyclable, Hazardous, Infectious
- Physical, Chemical, Biological Properties

CLASSIFICATIONS

1. **Industrial solid wastes**
   Assessment of industrial solid waste management problem greatly varies depending on the nature of the industry, their location and mode of disposal of waste. Further, for arriving at an appropriate solution for better management of industrial solid waste, assessment of nature of waste generated is also essential. Industries are required to collect and dispose of their waste at specific disposal sites and such collection, treatment and disposal is required to be monitored by the concerned State Pollution Control Board (SPCB) or Pollution Control Committee (PCC) in Union Territory.

   Problems
   The following problems are generally encountered in cities and towns while dealing with industrial solid waste:
   - There are no specific disposal sites where industries can dispose their waste;
   - Mostly, industries generating solid waste in city and town limits are of small-scale nature and even do not seek consents of SPCBs/PCCs
   - Industries are located in non-conforming areas and as a result they cause water and air pollution problems besides disposing solid waste.
   - Industrial estates located in city limits do not have adequate facilities so that industries can organise their collection, treatment and disposal of liquid and solid waste;
   - There is no regular interaction between urban local bodies and SPCBs/PCCs to deal

such issues relating to treatment and disposal of waste and issuance of licenses in non-conforming areas.

The major generators of industrial solid wastes are the thermal power plants producing coal ash, the integrated Iron and Steel mills producing blast furnace slag and steel melting slag, non-ferrous industries like aluminium, zinc and copper producing red mud and tailings, sugar industries generating press mud, pulp and paper industries producing lime and fertilizer and allied industries producing gypsum.

a) Coal Ash
Coal ash, also referred to as coal combustion residuals or CCRs, and is produced primarily from the burning of coal in coal-fired power plants. Indigenous technology for construction of building materials utilizing fly ash is available and are being practiced in a few industries. However, large scale utilization is yet to take off. Even if the full potential of fly ash utilization through manufacture of fly ash bricks and blocks is explored, the quantity of fly ash produced by the thermal power plants are so huge that major portion of it will still remain unutilized. Hence, there is a need to evolve strategies and plans for safe and environmentally sound method of disposal.

b) Integrated Iron & Steel Plant Slag
The Blast Furnace (BF) and Steel Melting Shop (SMS) slags in integrated iron and steel plants are at present dumped in the surrounding areas of the steel plants making hillocks encroaching on the agricultural land. Although, the BF slag has potential for conversion into granulated slag, which is a useful raw material in cement manufacturing, it is yet to be practiced in a big way. Even the use of slag as road subgrade or land-filling is also very limited.

c) Phosphogypsum
Phosphogypsum is the waste generated from the phosphoric acid, ammonium phosphate and hydrofluoric acid plants. This is very useful as a building material. At present very little attention has been paid to its utilization in making cement, gypsum board, partition panel, ceiling tiles, artificial marble, fiber boards etc.

d) Red Mud
Red mud as solid waste is generated in non-ferrous metal extraction industries like aluminium and copper. The red mud at present is disposed in tailing ponds for settling, which more often than not finds its course into the rivers, especially during monsoon. However, red mud has recently been successfully tried and a plant has been set up in the country for making corrugated sheets. Attempts are also made to manufacture polymer and natural fibres composite panel doors from red mud.

e) Lime Mud
Lime sludge, also known as lime mud, is generated in pulp & paper mills which is not recovered for reclamation of calcium oxide for use except in the large mills. The lime mud disposal by dumping into low-lying areas or into water courses directly or as run-off during monsoon is not only creating serious pollution problem but also wasting the valuable non- renewable resources.

Collection and transport

Manual handling of industrial waste is the usual practice in developing countries; there are very few mechanical aids for waste management. Wastes are shoveled by hand into storage containers and loaded manually into lorries. The people undertaking salvaging do so mainly by hand, picking out useful items, usually not even wearing gloves.

Apart from the likelihood of cuts caused by broken glass or sharp metals, sorting through waste contaminated with hazardous chemical materials could cause skin burns, excessive lacrimation, or even loss of consciousness; chronic hazards include respiratory problems from dust inhalation, and potential carcinogenicity from toxic chemicals present in discarded containers or surface deposits in other waste.

Personnel handling hazardous wastes should wear appropriate protective clothing. Mechanical methods for handling waste should be adopted wherever possible, and people should be educated about the dangers of manual handling of hazardous waste.

2. **Nuclear solid waste**
Solid radioactive wastes are produced from many systems and purification circuits of the reactor station. Low level waste can also sometimes be "declassified" and disposed of as normal, inactive, waste or scrap. Combustible solids may be reduced in volume through incineration.

Treatment of nuclear solid wastes

- Packaging: Packaging of solid radioactive waste by the waste generator for handling, transportation and further waste processing is an important pre-treatment operation.
- It has to comply with transport regulations (if transportation is involved), acceptance criteria or waste specifications for further waste processing and general occupational radiation protection standards.
- Combustible low-level solid waste is normally collected at the point of generation in transparent plastic bags (polyethylene or PVC) and is marked with the radiation symbol.
- This primary package is generally adapted to the collection system, which may be pedal bins for laboratories and other small waste generators or larger bins at nuclear facilities.
- After filling, the plastic bags are removed from the bins and closed with adhesive tape.
- Non-combustible small size low and intermediate level wastes are usually collected as compressible and non-compressible materials in 20–50 L metal or cardboard boxes for small generators and 100–200 L metal drums for larger generators.
- Standard high efficiency particulate air filters are often packed in welded plastic bags, especially when alpha contamination is present.
- Cardboard boxes may be used as an over-pack.
- Animal carcasses or similar types of waste may be packed in plastic bags and deep frozen to allow convenient accumulation and storage.

3. ELECTRONIC WASTE

- Electronic waste from equipment of all sizes includes dangerous chemicals like lead, cadmium, beryllium, mercury, and brominated flame retardants.
- When e-waste sits in a typical landfill, for example, water flows through the landfill and picks up trace elements from these dangerous minerals. Eventually the contaminated landfill water, called "leachate," gets through layers of natural and manufactured landfill liner and other protection. When it reaches natural groundwater, it introduces lethal toxicity.
- Health risks range from kidney disease and brain damage to genetic mutations.
- Proper or formal e-waste recycling usually involves disassembling the electronics, separating and categorizing the contents by material and cleaning them.
- Items are then shredded mechanically for further sorting with advanced separation technologies. Companies must adhere to health and safety rules and use pollution-control technologies that reduce the health and environmental hazards of handling e-waste. All this makes formal recycling expensive.

SOLID WASTE MANAGEMENT

Waste Management means

- Storage
- Collection
- Transport and handling
- Recycling
- Disposal and monitoring of waste materials.

Hazardous Waste

- Fatal to humans or other organisms at low quantities.
- Toxic, carcinogenic, mutagenic or teratogenic to humans or other organisms.
- Ignitable with a flash point less than 60 degree Celsius.
- Corrosive, explosive or highly chemically reactive.

CONTROL MEASURES

Solid waste management is a manifold task involving many activities like:

1. Collection of solid wastes.
2. Disposal of solid wastes.
3. Waste utilisation.

(i) Collection of Solid Wastes:

Collection includes all the activities associated with the gathering of solid wastes and the hauling of the wastes collected to the location from where the collection vehicle will ultimately transport it to the site of disposal. There are three basic methods of collection.

(a) Community storage point:
The municipal refuse is taken to fixed storage bins and stored till the waste collection agency collects it daily for disposal in a vehicle.

(b) Kerbside Collection:
In advance of the collection time, the refuse is brought in containers and placed on the footway from where it is collected by the waste collection agency.

(c) Block Collection:
Individuals bring the waste in containers and hand it over to the collection staff who empties it into the waiting vehicle and returns the container to the individuals.

(ii) Disposal of Solid Wastes:

Due to heterogeneity of the city refuse it is important to select the most appropriate solid waste disposal method keeping in view the following objectives:

(a) It should be economically viable i.e. the operation and maintenance costs must be carefully assessed.
(b) It should not create a health hazard.
(c) It should not cause adverse environmental effects.
(d) It should not be aesthetically unpleasant i.e. it should not result in offending sights, odours, and noises.
(e) It should preferably provide opportunities for recycling of materials.

Incineration

It is the process of burning municipal solid waste in a properly designed furnace under suitable temperature and operating conditions. It reduces the municipal solid waste by about 90% and 75% by weight.

Composting

Bacterial decomposition of organic components of the municipal waste result in the formation of humus or compost and the process is known as composting.

It helps in disposal of solid waste, disposal of night soil, and production of valuable manure for crops, it is also termed as biodegradation.

Recycling

It means reusing some components of the waste that may have some economic value. Recycling conserves resources, reduce the energy used during manufacture and also reduce pollution.

Source recovery

It is a kind of destructive distillation in which the solid wastes are heated in pyrolysis reactor at 650-1000 degree centigrade in oxygen depleted environment. By this process, the chemical constituents and chemical energy of some organic wastes are recovered.

The organic constituents split up into gaseous liquid and gaseous fractions like carbon dioxide, carbon monoxide, tar, methane, charred carbon etc.

Source reduction

It is one of the fundamental ways to reduce waste. This can be done by using less material when making a product, reusing products, designing products packaging to reduce their quantity. Individually one can reduce the use of unnecessary items which causes solid waste.

(iii) Waste Utilisation:

A developing country cannot afford wastage. By proper utilisation of solid waste, a developing country like India can avail of many advantages, for instance:

(a) Waste utilisation directly or indirectly contributes to economic development.
(b) Waste utilisation generates employment opportunities.
(c) Unused solid wastes create environmental hazards by spreading diseases and causing air and water pollution.
(d) Waste utilisation helps in conservation of natural resources.

Waste utilisation helps to generate many useful products which are the basic necessities of life

CONCLUSION

- In general, disposal practices for solid and hazardous waste products have not been satisfactory.
- One of the most important solutions to this problem is to produce less waste.
- When compared to Japan, Canada and the United States produce about two or three times more waste per person despite having similar levels of economic prosperity.

REDUCE REUSE RECYCLE

- Large quantities of municipal waste are generated from consumer product packaging.
- To combat this problem, recyclable containers or containers with considerably less packaging

was introduced.

- Another strategy used to reduce waste is wrap products in biodegradable containers.
- Reusing the containers also reduce production of wastes.

GLOBAL WARMING

Global warming is the slow increase in the average temperature of the earth's atmosphere because an increased amount of the energy (heat) striking the earth from the sun is being trapped in the atmosphere and not radiated out into space. Earth's average temperature has been steadily rising ever since humans started burning fossil fuels following Industrial Revolution leading to a rise in the level of carbon dioxide and other greenhouse gases in the atmosphere. Since the start of the Industrial Revolution, the Earth's average temperature has steadily risen. Human-induced warming reached approximately 1°C (likely between 0.8°C and 1.2°C) above pre-industrial levels in 2017, increasing at 0.2°C (likely between 0.1°C and 0.3°C) per decade (high confidence). [The pre-industrial levels are the baseline when the scientist assume that the effect of human-induced climate change was theoretically zero.] If these activities continue unchecked, there is a possibility that the temperature would cross 1.5°C mark and even breach 2°C. Global warming is the cause, climate change is the effect. Scientists often prefer to speak about climate change instead of global warming, because higher global temperatures don't necessarily mean that it will be warmer at any given time at every location on Earth. Warming is strongest at the Earth's Poles, the Arctic and the Antarctic, and will continue to be so.

Climate change

The term climate refers to the general weather conditions of a place over many years. In the United States, for example, Maine's climate is cold and snowy in winter while South Florida's is tropical year-round. Climate change is a significant variation of average weather conditions—say, conditions becoming warmer, wetter, or drier—over several decades or more. It's that longer-term trend that differentiates climate change from natural weather variability. And while "climate change" and "global warming" are often used interchangeably, global warming—the recent rise in the global average temperature near the earth's surface —is just one aspect of climate change.

Other effects of global warming

- Rising sea levels: Climate change impacts rising sea levels. Average sea level around the world rose about 8 inches (20 cm) in the past 100 years; climate scientists expect it to rise more and more rapidly in the next 100 years as part of climate change impacts.
- Melting ice: Projections suggest climate change impacts within the next 100 years, if not sooner, the world's glaciers will have disappeared, as will the Polar ice cap, and the huge Antarctic ice shelf, Greenland may be green again, and snow will have become a rare phenomenon at what are now the world's most popular ski resorts.
- Forests, farms, and cities will face troublesome new pests, heat waves, heavy downpours, and increased flooding. All those factors will damage or destroy agriculture and fisheries.
- Disruption of habitats such as coral reefs and Alpine meadows could drive many plant and animal species to extinction.
- Allergies, asthma, and infectious disease outbreaks will become more common due to increased growth of pollen-producing ragweed, higher levels of air pollution, and the spread of conditions favourable to pathogens and mosquitoes.

The greenhouse effect

The exchange of incoming and outgoing radiation that warms the Earth is often referred to as the greenhouse effect because a greenhouse works in much the same way. Incoming UV radiation easily passes through the glass walls of a greenhouse and is absorbed by the plants and hard surfaces inside. Weaker IR radiation, however, has difficulty passing through the glass walls and is trapped inside, thus warming the greenhouse. This effect lets tropical plants thrive inside a greenhouse, even during a cold winter. A similar phenomenon takes place in a car parked outside on a cold, sunny day. Incoming solar radiation warms the car's interior, but outgoing thermal radiation is trapped inside the car's closed windows.

Gases that contribute to the greenhouse effect include:

- **Water vapour**: The most abundant greenhouse gas, but importantly, it acts as a feedback to the climate. Water vapor increases as the Earth's atmosphere warms, but so does the possibility of clouds and precipitation, making these some of the most important feedback mechanisms to the greenhouse effect. Carbon dioxide (CO_2). A minor but very important component of the atmosphere, carbon dioxide is released through natural processes such as respiration and volcano eruptions and through human activities such as deforestation, land use changes, and burning fossil fuels. Humans have increased atmospheric CO_2 concentration by more than a third since the Industrial Revolution began. This is the most important long-lived "forcing" of climate change.
- **Methane**. A hydrocarbon gas produced both through natural sources and human activities, including the decomposition of wastes in landfills, agriculture, and especially rice cultivation, as well as ruminant digestion and manure management associated with domestic livestock. On a molecule-for-molecule basis, methane is a far more active greenhouse gas than carbon dioxide, but also one which is much less abundant in the atmosphere.
- **Nitrous oxide**. A powerful greenhouse gas produced by soil cultivation practices, especially the use of commercial and organic fertilizers, fossil fuel combustion, nitric acid production, and biomass burning.
- **Chlorofluorocarbons (CFCs).** Synthetic compounds entirely of industrial origin used in a number of applications, but now largely regulated in production and release to the atmosphere by international agreement for their ability to contribute to destruction of the ozone layer. They are also greenhouse gases.

Sources of emission Global greenhouse gas emissions can also be broken down by the economic activities that lead to their production

- Electricity and Heat Production (25% of 2010 global greenhouse gas emissions): The burning of coal, natural gas, and oil for electricity and heat is the largest single source of global greenhouse gas emissions.
- Industry (21% of 2010 global greenhouse gas emissions): Greenhouse gas emissions from industry primarily involve fossil fuels burned on site at facilities for energy. This sector also includes emissions from chemical, metallurgical, and mineral transformation processes not associated with energy consumption and emissions from waste management activities. (Note: Emissions from industrial electricity use are excluded and are instead covered in the Electricity and Heat Production sector.)
- Agriculture, Forestry, and Other Land Use (24% of 2010 global greenhouse gas emissions): Greenhouse gas emissions from this sector come mostly from agriculture (cultivation of crops and livestock) and deforestation. This estimate does not include the CO_2 that ecosystems remove from the atmosphere by sequestering carbon in biomass, dead organic matter, and soils, which offset approximately 20% of emissions from this sector.
- Transportation (14% of 2010 global greenhouse gas emissions): Greenhouse gas emissions from this sector primarily involve fossil fuels burned for road, rail, air, and marine transportation. Almost all (95%) of the world's transportation energy comes from petroleum-based fuels, largely gasoline and diesel.
- Buildings (6% of 2010 global greenhouse gas emissions): Greenhouse gas emissions from this sector arise from onsite energy generation and burning fuels for heat in buildings or cooking in homes. (Note: Emissions from electricity use in buildings are excluded and are instead covered in the Electricity and Heat Production sector.)
- Other Energy (10% of 2010 global greenhouse gas emissions): This source of greenhouse gas emissions refers to all emissions from the Energy sector which are not directly associated with electricity or heat production, such as fuel extraction, refining, processing, and transportation.

Cases

The current warming trend is of particular significance because most of it is extremely likely (greater than 95 percent probability) to be the result of human activity since the mid-20th century and proceeding at a rate that is unprecedented over decades to millennia.

Global Temperature Rise: The planet's average surface temperature has risen about 1.62 degrees Fahrenheit (0.9 degrees Celsius) since the late 19th century, a change driven largely by increased carbon dioxide and other human-made emissions into the atmosphere. Most of the warming occurred in the past 35 years, with the five warmest years on record taking place since 2010. Not only was 2016 the warmest year on record, but eight of the 12 months that make up the year — from January through September, with the exception of June — were the warmest on record for those respective months.

Warming Oceans: The oceans have absorbed much of this increased heat, with the top 700 meters (about 2,300 feet) of ocean showing warming of more than 0.4 degrees Fahrenheit since 1969.

Shrinking Ice Sheets: The Greenland and Antarctic ice sheets have decreased in mass. Data from NASA's Gravity Recovery and Climate Experiment show Greenland lost an average of 286 billion tons of ice per year between 1993 and 2016, while Antarctica lost about 127 billion tons of ice per year during the same time period. The rate of Antarctica ice mass loss has tripled in the last decade

Glacial Retreat: Glaciers are retreating almost everywhere around the world including in the Alps, Himalayas, Andes, Rockies, Alaska and Africa.

Decreased Snow Cover: Satellite observations reveal that the amount of spring snow cover in the Northern Hemisphere has decreased over the past five decades and that the snow is melting earlier.

Sea Level Rise: Global sea level rose about 8 inches in the last century. The rate in the last two decades, however, is nearly double that of the last century and is accelerating slightly every year.

Declining Arctic Sea Ice: Both the extent and thickness of Arctic sea ice has declined rapidly over the last several decades.

Ocean Acidification: Since the beginning of the Industrial Revolution, the acidity of surface ocean waters has increased by about 30 percent. This increase is the result of humans emitting more carbon dioxide into the atmosphere and hence more being absorbed into the oceans. The amount of carbon dioxide absorbed by the upper layer of the oceans is increasing by about 2 billion tons per year.

Solutions to global warming and climate change

- **Renewable energies:** The first way to prevent climate change is to move away from fossil fuels and using alternative methods such as Renewable energies like solar, wind, biomass and geothermal.
- **Energy & water efficiency:** Producing clean energy is essential, but reducing our consumption of energy and water by using more efficient devices (e.g. LED light bulbs, innovative shower system) is less costly and equally important.
- **Sustainable transportation:** Promoting public transportation, carpooling, but also electric and hydrogen mobility, can definitely help reduce CO_2 emissions and thus fight global warming.
- **Sustainable infrastructure:** In order to reduce the CO_2 emissions from buildings - caused by heating, air conditioning, hot water or lighting - it is necessary both to build new low energy buildings, and to renovate the existing constructions.
- **Sustainable agriculture & forest management:** Encouraging better use of natural resources, stopping massive deforestation as well as making agriculture greener and more efficient should also be a priority.
- **Responsible consumption & recycling:** Adopting responsible consumption habits is crucial, be it regarding food (particularly meat), clothing, cosmetics or cleaning products.

  Last but not least, recycling is an absolute necessity for dealing with waste.

Indian Govt. Policies:

Various legislations have been enacted by Indian Parliament in about the last 30 years to tackle the problem of environmental protection.

- In terms of Article 21 of the Constitution, a person has a right to a decent life, good environment and maintenance of ecology. Therefore, when we talk of environment degradation, we talk of violation of rights under Article 21.The State's responsibility with regard to environmental protection has been laid down under Article 48-A of our Constitution, which reads as follows: "The State shall endeavour to protect and improve the environment and to safeguard the forests and wildlife of the country" Environmental protection is a fundamental duty of every citizen of this country under Article 51-A(g) of our Constitution which reads as follows: "It shall be the duty of every citizen of India to protect and improve the natural environment including forests, lakes, rivers and wildlife and to have compassion for living creatures."
- The environment and human life are interlinked hence the realization of many human rights are necessarily related to and in some ways dependent upon one's physical environment (as per the Inter-American Commission on Human Rights).

CO_2 EMISSIONS, AQI AND GOVERNMENT REGULATIONS

Sources of CO_2 Emissions

Sources of CO_2 Emissions are categorised into 2 types: Natural and Human sources

Carbon Dioxide Emissions: Human Sources

87 percent of all human-produced carbon dioxide emissions come from the burning of fossil fuels like coal, natural gas and oil. The remainder results from the clearing of forests and other land use changes (9%), as well as some industrial processes such as cement manufacturing (4%).

- **Fossil fuel combustion/use**
  The largest human source of carbon dioxide emissions is from the combustion of fossil fuels. This produces 87% of human carbon dioxide emissions. Burning these fuels releases energy, which is most commonly turned into heat, electricity or power for transportation. Some examples of where they are used are in power plants, cars, planes and industrial facilities. In 2011, fossil fuel use created 33.2 billion tonnes of carbon dioxide emissions worldwide. The 3 types of fossil fuels that are used the most are coal, natural gas and oil. Coal is responsible for 43% of carbon dioxide emissions from fuel combustion, 36% is produced by oil and 20% from natural gas. Coal is the most carbon intensive fossil fuel. The three main economic sectors that use fossil fuels are: electricity/heat, transportation and industry. The first two sectors, electricity/heat and transportation, produced nearly two-thirds of global carbon dioxide emissions in 2010.
  - **Electricity and heat generation** is the economic sector that produces the largest amount of man-made carbon dioxide emissions. This sector produced 41% of fossil fuel related carbon dioxide emissions in 2010. Around the world, this sector relies heavily on coal, the most carbon-intensive of fossil fuels, explaining this sector giant carbon footprint. Almost all industrialized nations get the majority of their electricity from the combustion of fossil fuels (around 60-90%). Only Canada and France are the exception. Depending on the energy mix of your local power company you probably will find that the electricity that you use at home and at work has a considerable impact on greenhouse gas emissions.
  - **The transportation sector** is the second largest source of anthropogenic carbon dioxide emissions. Transporting goods and people around the world produced 22% of fossil fuel related carbon dioxide emissions in 2010. This sector is very energy intensive and it uses petroleum-based fuels (gasoline, diesel, kerosene, etc.) almost exclusively to meet those needs. Since the 1990s, transport related emissions have grown rapidly, increasing by 45% in less than 2 decades. Road transport accounts for 72% of this sector's carbon dioxide emissions. Automobiles, freight and light-duty trucks are the main sources of emissions for the whole transport sector and emissions from these three have steadily grown since 1990. Apart from road vehicles, the other important sources of emissions for this sector are marine shipping and global aviation.
  - **The industrial sector** is the third largest source of man-made carbon dioxide emissions. This sector produced 20% of fossil fuel related carbon dioxide emissions in 2010. The industrial sector consists of manufacturing, construction, mining, and agriculture. Manufacturing is the largest of the 4 and can be broken down into 5 main categories: paper, food, petroleum refineries, chemicals, and metal/mineral products. These categories account for the vast majority of the fossil fuel use and CO2 emissions by this sector. Manufacturing and industrial processes all combine to produce large amounts of each type of greenhouse gas but specifically large amounts of CO2. This is because many

manufacturing facilities directly use fossil fuels to create heat and steam needed at various stages of production. For example, factories in the cement industry, have to heat up limestone to 1450°C to turn it into cement, which is done by burning fossil fuels to create the required heat.

- **Land use changes**
  Land use changes are a substantial source of carbon dioxide emissions globally, accounting for 9% of human carbon dioxide emissions and contributed 3.3 billion tonnes of carbon dioxide emissions in 2011. Land use changes are when the natural environment is converted into areas for human use like agricultural land or settlements. From 1850 to 2000, land use and land use change released an estimated 396-690 billion tonnes of carbon dioxide to the atmosphere, or about 28-40% of total anthropogenic carbon dioxide emissions. Trees act as a carbon sink. They remove carbon dioxide from the atmosphere via photosynthesis. When forests are cleared to create farms or pastures, trees are cut down and either burnt or left to rot, which adds carbon dioxide to the atmosphere.

Carbon Dioxide Emissions: Natural Sources

42.84 percent of all naturally produced carbon dioxide emissions come from ocean-atmosphere exchange. Other important natural sources include plant and animal respiration (28.56%) as well as soil respiration and decomposition (28.56%). A minor amount is also created by volcanic eruptions (0.03%).

Air Quality Index (AQI)

The air quality index (AQI) is an index for reporting air quality on a daily basis. It is a measure of how air pollution affects one's health within a short time period. The purpose of the AQI is to help people know how the local air quality impacts their health. The US Environmental Protection Agency (EPA) calculates the AQI for five major air pollutants, for which national air quality standards have been established to safeguard public health.

a) Ground-level ozone
b) Particle pollution/particulate matter (PM2.5/pm 10)
c) Carbon Monoxide
d) Sulphur dioxide
e) Nitrogen dioxide

The higher the AQI value, the greater the level of air pollution and the greater the health concerns.

Computing the AQI

The air quality index is a piecewise linear function of the pollutant concentration. At the boundary between AQI categories, there is a discontinuous jump of one AQI unit. To convert from concentration to AQI this equation is used:

$$I = \frac{I_{high} - I_{low}}{C_{high} - C_{low}}(C - C_{low}) + I_{low}$$

where,

I = the (Air Quality) index,
C = the pollutant concentration,
C_{low} = the concentration breakpoint that is ≤ C,
C_{high} = the concentration breakpoint that is ≥ C,
I_{low} = the index breakpoint corresponding to C_{low},
I_{high} = the index breakpoint corresponding to C_{high}.

Air Quality Index Levels of Health Concern	Numerical Value	Meaning
Good	0 to 50	Air quality is considered satisfactory, and air pollution poses little or no risk.
Moderate	51 to 100	Air quality is acceptable; however, for some pollutants there may be a moderate health concern for a very small number of people who are unusually sensitive to air pollution.
Unhealthy for Sensitive Groups	101 to 150	Members of sensitive groups may experience health effects. The general public is not likely to be affected.
Unhealthy	151 to 200	Everyone may begin to experience health effects; members of sensitive groups may experience more serious health effects.
Very Unhealthy	201 to 300	Health alert: everyone may experience more serious health effects.
Hazardous	301 to 500	Health warnings of emergency conditions. The entire population is more likely to be affected.

Government regulations in controlling Pollution

Some of the important legislations for environment protection are as follows:

- The National Green Tribunal Act, 2010
- The Air (Prevention and Control of Pollution) Act, 1981
- The Water (Prevention and Control of Pollution) Act, 1974
- The Environment Protection Act, 1986
- The Hazardous Waste Management Regulations, etc.

The National Green Tribunal Act, 2010

The National Green Tribunal Act, 2010 (No. 19 of 2010) (NGT Act) has been enacted with the objectives to provide for establishment of a National Green Tribunal (NGT) for the effective and expeditious disposal of cases relating to environment protection and conservation of forests and other natural resources including enforcement of any legal right relating to environment and giving relief and compensation for damages to persons and property and for matters connected therewith or incidental thereto.

The Air (Prevention and Control of Pollution) Act, 1981

The Air (Prevention and Control of Pollution) Act, 1981 (the "Air Act") is an act to provide for the prevention, control and abatement of air pollution and for the establishment of Boards at the Central and State levels with a view to carrying out the aforesaid purposes.

To counter the problems associated with air pollution, ambient air quality standards were established under the Air Act. The Air Act seeks to combat air pollution by prohibiting the use of polluting fuels and substances, as well as by regulating appliances that give rise to air pollution.

The Water (Prevention and Control of Pollution) Act, 1974

The Water Prevention and Control of Pollution Act, 1974 (the "Water Act") has been enacted to provide for the prevention and control of water pollution and to maintain or restore wholesomeness of water in the country. It further provides for the establishment of Boards for the prevention and control of water pollution with a view to carry out the aforesaid purposes. The Water Act prohibits the discharge of pollutants into water bodies beyond a given standard, and lays down penalties for non-compliance.

The Environment Protection Act, 1986

The Environment Protection Act, 1986 (the "Environment Act") provides for the protection and improvement of the environment. The Environment Protection Act establishes the framework for studying, planning and implementing long-term requirements of environmental safety and laying down a system of speedy and adequate response to situations threatening the environment. It is an umbrella legislation designed to provide a framework for the coordination of central and state authorities established under the Water Act, 1974 and the Air Act.

Hazardous Wastes Management Regulations

Hazardous waste means any waste which, by reason of any of its physical, chemical, reactive, toxic, flammable, explosive or corrosive characteristics, causes danger or is likely to cause danger to health or environment, whether alone or when in contact with other wastes or substances.

There are several legislations that directly or indirectly deal with hazardous waste management. The relevant legislations are the Factories Act, 1948, the Public Liability Insurance Act, 1991, the National Environment Tribunal Act, 1995 and rules and notifications under the Environmental Act.

EMISSION FACTOR

Introduction

Emissions factors have long been the fundamental tool in developing national, regional, state, and local emissions inventories for air quality management decisions and in developing emissions control strategies. More recently, emissions factors have been applied in determining site-specific applicability and emissions limitations in operating permits by federal, state, local, and tribal agencies, consultants, and industry. These users have requested guidance on the use of emissions factors and other emissions quantification tools (e.g, emissions testing and monitoring, mass balance techniques) in developing permits that are practical in their enforcement.

An emission factor is a representative value that attempts to relate the quantity of a pollutant released to the atmosphere with an activity associated with the release of that pollutant. These factors are usually expressed as the weight of pollutant divided by a unit weight, volume, distance, or duration of the activity emitting the pollutant (e.g., kilograms of particulate emitted per megagram of coal burned). Such factors facilitate estimation of emissions from various sources of air pollution. In most cases, these factors are simply average of all available data of acceptable quality, and are generally assumed to be representative of long-term averages for all facilities in the source category (i.e., a population average).

Definition

An emissions factor is a representative value that attempts to relate the quantity of a pollutant released to the atmosphere with an activity associated with the release of that pollutant. These factors are usually expressed as the weight of pollutant divided by a unit weight, volume, distance, or duration of the activity emitting the pollutant (e.g., kilograms of particulate emitted per megagram of coal burned). Such factors facilitate estimation of emissions from various sources of air pollution. In most cases, these factors are simply averages of all available data of acceptable quality, and are generally assumed to be representative of long-term averages for all facilities in the source category (i.e., a population average).The general equation for emissions estimation is:

$$E = A \times EF \times (1 - ER/1\,00)$$

where:

E = emissions;

A = activity rate;

EF = emission factor, and

ER =overall emission reduction efficiency, %

An emission factor is a coefficient which allows to convert activity data into GHG emissions. It is the average emission rate of a given source, relative to units of activity or process/processes.

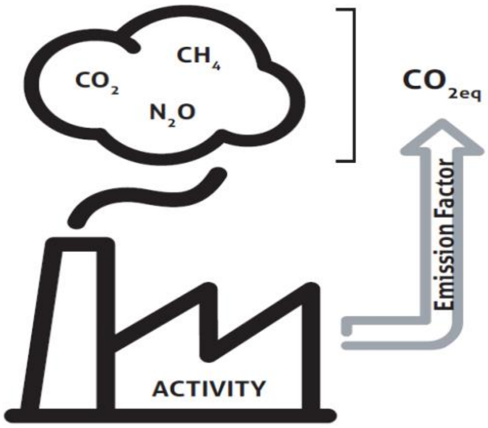
For example: the natural gas emits 0.244 kg CO_2eq / kWh ICV (European mean) with 5% uncertainty.

Thus, the emission factor is the sum of emissions of $CO_2eq$of the human activity described as mass unit of CO_2eq / reference flows.

For example: the EF for the natural gas is the sum of the combustion (0.205 kg CO2eq/ kWh ICV) and the upstream (i.e. the production and transport of the gas) (0.0389 kg CO2eq / kWh ICV).

<u>CO2eq</u>

A human activity emits different kind of greenhouse gas (GHG) (see below). Their Global Warming Potential (GWP), a physical characteristic of a GHG, represents their impact on the greenhouse effect, and allows to convert 1 kg of GHG into X kg of CO2 equivalent, noted CO2eq. Thus, emissions of different gases can be compared.



Fossil carbon dioxide	Biogenic carbon dioxide	Methane	Nitrous oxide	Hydro-fluoro-carbons	Perfluoro-carbons	Sulfur hexafluoride	Nitrogen trifluoride
CO_2	CO_{2b}	CH_4	N_2O	HFCs	PFCs	SF_6	NF_3

<u>Uncertainty</u>

In the countries where data are not specific, organisations are reluctant to apply international values since the accuracy of the results is not guaranteed (due to difference between the localisations). Indeed, the calculation of carbon footprints is then based on generic data that can be totally different to the real situation, and thus have a high uncertainty. This has major impact on the credibility of any results coming from any calculation method.

When developing an emission factor, the issue of uncertainty and its evaluation arises. Moreover, different sources of uncertainty can be identified:

- **Parameter uncertainty:**
  A measure of how close the data used to calculate emissions are to the actual data and real emissions. For example: GWP values are associated with an uncertainty of ± 35% for the 90% confidence interval. The estimated uncertainty of emissions from individual sources (e.g. power plants, motor vehicles, dairy cattle) is either a function of instrument characteristics, calibration and sampling frequency of direct measurements, or (more often) a combination of the uncertainties in the emission factors for typical sources and the corresponding activity data.
- **Model uncertainty:**
  Limitations in the ability of the modelling approach used to reflect the real world;
- **Scenario uncertainty:** methodological choices allocation, product use assumption, EoL assumptions.